

Performance Testing

Date	15 March 2026
Team ID	UNM1659GenieBirds
Project Name	EduGenie – AI-Powered Learning Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the EduGenie system behaves under expected and peak load conditions. The following document records the planned test approach, scenarios, target metrics and observations. Actual tool-generated evidence should be attached where indicated.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter / k6 for API load and response-time testing
Type of Testing	Load Testing, Stress Testing and Basic Spike Testing
Target Module / API	EduGenie question-answering / AI response API and quiz-generation request flow
Test Environment	Local development / hosted API test environment
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Normal user load – submit questions and receive AI answers	10	60	Stable responses with no failed requests
2	Moderate load – simultaneous topic explanations and summaries	25	120	Response time remains within the defined target
3	Peak load – concurrent quiz generation and question requests	50	180	System remains available with acceptable latency
4	Spike test – sudden increase in concurrent requests	100	60	No crash; service recovers after the spike

Note: The scenario values above are the planned test configuration. They should be executed in the selected performance-testing tool and the actual measurements should be entered on Page 2.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	To be measured	Pending	Record tool output
2	Response Time (Max)	< 5 seconds	To be measured	Pending	Record peak latency
3	Throughput (Req/sec)	Record baseline	To be measured	Pending	Compare across load levels
4	Error Rate	< 1%	To be measured	Pending	Check failed/timeout requests
5	CPU Utilization	< 80%	To be measured	Pending	Monitor server during test
6	Memory Utilization	< 80%	To be measured	Pending	Monitor memory usage

Step 4: Observations & Analysis

Key Findings:

Performance should be evaluated by comparing average and maximum response time, throughput and error rate at each load level. AI-generated responses may have higher latency than simple application requests.

Bottlenecks Identified:

Potential bottlenecks include AI/API response latency, network delay, concurrent request handling and server CPU/memory usage. Confirm the actual bottleneck from the performance-tool report before recording it as a finding.

Optimization Steps Taken:

Use request timeouts, efficient API calls, appropriate concurrency limits, response caching where applicable, and resource monitoring. Record only optimizations actually implemented in the project.

Step 5: Screenshots / Evidence

Attach screenshots of the performance-testing tool results below (for example, JMeter report, k6 output or graphs). No test-result screenshots were supplied with the template, so the evidence areas are intentionally left for the team's actual execution results.

Evidence 1 – Insert performance test report / graph here

Evidence 2 – Insert response-time / throughput / error-rate result here